# Supplementary Material for *Look Before You Look: Auditing Perception Utility and Evaluation Sensitivity in Multimodal Agents*

Anonymous Author(s)
AAMAS 2027 Submission

## Contents

**1 Visual-Token Budget Accounting** **2**

**2 Perception Actions: Exact Specifications** **2**
2.1 Crop candidate construction ($K = 8$, constant cardinality) . . . 2
2.2 Control (dummy) actions and the luck floor . . . 4
2.3 OCR as an external tool . . . 4
2.4 HighRes and Stop . . . 4

**3 Prompts (Verbatim)** **4**

**4 Replay Construction** **5**

**5 Scoring Rules** **6**

**6 Diagnostic and Selector Features** **6**

**7 Frozen Policies, Baselines, and Artifacts** **7**

**8 Preregistration Index and Deviations Log** **8**

**9 Additional Results** **10**
9.1 Budget-constrained oracle frontier (Gate-0 logs, Qwen) . . . 10
9.2 Exact constrained oracle vs. Lagrangian relaxation . . . 11
9.3 Per-action damage decomposition . . . 11
9.4 Answer-span confidence variants . . . 11
9.5 Second reader (Phi-4): lenient rescoring and realizable signal . . . 11
9.6 Multi-replay robustness (replays 3–5) . . . 11
9.7 Randomized-mixture baselines on the held-out sets . . . 12
9.8 Prospective one-step evaluation: realized-cost matching . . . 12
9.9 Sequential experiments: full cell table . . . 12
9.10 Hidden-state selector v3: budget-control failure . . . 12
9.11 F0 nonlinear-head prospective follow-up . . . 13
9.12 Representation–head $3 \times 3$ prospective test . . . 13
9.13 Action-matched counterfactual oracle . . . 13
9.14 Blinded human adjudication of scorer disagreement . . . 14
9.15 Cross-dataset and cross-reader transfer details . . . 15
9.16 Latency at the token-matched operating points . . . 15

**10 Infrastructure and Reproducibility Details** **16**

# Overview

This document provides implementation details deferred from the main paper (exact crop coordinates, prompts, filtering rules, fallbacks, OCR, and scoring details; Sections 1–6), frozen policy and controller specifications (Section 7), the preregistration index and deviations log (Section 8), and additional analyses supporting the paper's three-level distinction among unrestricted structural opportunity, action-matched counterfactual opportunity, and frozen prospective policy performance (Section 9). Section 9 also reports the blinded human-adjudication analysis and the nonlinear and representation–head prospective follow-ups. Sections 10–11 document infrastructure, data separation, and the accompanying reproducibility artifacts.

All automatically scored quantities are recomputable offline from the released result files because raw generations are preserved for every rollout. Human adjudication is provided separately through the frozen annotation sample, blinded labels, and adjudication summaries.

# 1 Visual-Token Budget Accounting

Visual cost is measured in Qwen2.5-VL visual tokens, where one token covers a $28 \times 28$ pixel block after smart resizing (14-pixel patches with $2 \times 2$ merging):

$$\text{tokens}(h, w) = \left\lfloor \bar{h}/28 \right\rfloor \cdot \left\lfloor \bar{w}/28 \right\rfloor, \qquad (\bar{h}, \bar{w}) = \text{smart_resize}(h, w),$$

where smart_resize mirrors the official `qwen_vl_utils.smart_resize`: aspect ratio is preserved, both sides are snapped to multiples of 28, and the total area is clamped to [min_pixels, max_pixels] with max_pixels $= B \cdot 28^2$ for a budget of $B$ tokens (code: `ppu/imaging.py`).

We resize images ourselves and transmit the resized bytes (JPEG, quality 92, no chroma subsampling) rather than passing server-side pixel limits, so the realized budget is identical across backends and replays and is auditable from the logs. Every rendered view is logged together with its realized token count and a SHA-1 hash of its pixels, which lets the audit verify that distinct crop candidates are distinct observations.

**Per-dataset budgets.** Coarse / HighRes visual-token targets are 256/1,280 for MME-RealWorld-Lite, 192/512 for TextVQA, and 64/192 for VQAv2. Crop views are re-encoded at 256 tokens on all datasets. The full-information reference view (used only for diagnostics, never as a policy action) is capped at 2,560 tokens. A sample enters the audit only if its native view costs at least 4× the coarse budget; otherwise Crop and HighRes cannot add information (inclusion rule).

Views below 4 tokens are rejected as unreadable (min_pixels $= 4 \cdot 28^2$), and crop boxes that map to fewer than 8 pixels on a side raise an error rather than silently degrading.

# 2 Perception Actions: Exact Specifications

## 2.1 Crop candidate construction ($K = 8$, constant cardinality)

Every evaluated state has exactly eight crop candidates, but the construction depends on the experimental phase. Constant cardinality is important because the unrestricted oracle maximizes over crop candidates; varying the number of candidates would change the amount of selection-on-noise opportunity.

**Initial structural audit.** The initial structural audit (Gate-0, both readers) uses five deterministic multiscale grid boxes together with up to three frozen question-conditioned proposals. Proposal generation, parsing, filtering, and deterministic padding follow the procedure described below.

**Held-out and prospective collections.** All held-out and prospective collections remove VLM-generated proposals and use the same eight fixed boxes for every image: the five grid boxes below plus three smaller deterministic fallback boxes. Thus the deployed selectors, the state-independent mixtures, and the action-matched oracle operate over an identical fixed crop family.

**Deterministic multiscale grid (5 boxes).** Normalized $(x_0, y_0, x_1, y_1)$ coordinates:

| Region | Box |
|---|---|
| Center | (0.25, 0.25, 0.75, 0.75) |
| Top-left | (0.00, 0.00, 0.55, 0.55) |
| Top-right | (0.45, 0.00, 1.00, 0.55) |
| Bottom-left | (0.00, 0.45, 0.55, 1.00) |
| Bottom-right | (0.45, 0.45, 1.00, 1.00) |

The sequential experiment's frozen $\text{CROP}_{\text{TR}}$ action is the top-right grid box above; its first-step center crop is the center grid box.

**Fixed fallback boxes for held-out and prospective collections.** The three additional fixed boxes are

$$(0.35, 0.35, 0.65, 0.65), \quad (0.05, 0.05, 0.35, 0.35), \quad (0.65, 0.65, 0.95, 0.95),$$

deliberately smaller than the grid so that they probe a different scale. Together with the five grid boxes, these define the fixed eight-crop family used in all held-out and prospective one-step evaluations (crop budget 256 tokens each, coarse overview retained).

**Question-conditioned proposals in the initial structural audit only.** A single proposal generation is requested from the frozen reader with the prompt in Section 3. Its cost is charged to every arm that considers CROP at all, including random and oracle arms. Parsing is tolerant but never creative:

- Each output line must contain at least four numbers; lines that do not are dropped, not repaired.
- Qwen2.5-VL is grounded in absolute pixels and frequently mixes pixel and fractional coordinates in one line. Each coordinate is therefore interpreted independently: a value with $|v| \leq 1$ is read as a fraction, otherwise it is divided by the width or height of the view the model actually saw. Every parsed box logs its raw values, per-coordinate interpretation mode, and the reason for any rejection, so the decision is auditable per line. (An earlier all-fractional parser silently dropped $\sim$70% of proposal calls; the mixed parser and this failure are disclosed in the code comments.)
- Filters, applied in order: out-of-range after interpretation (tolerance $\pm 0.01$); degenerate or below 1% of image area; area above 0.95 (near-duplicate of the full view); IoU $> 0.9$ with an already accepted box. At most 3 proposals are kept.

**Padding to $K = 8$ in the initial structural audit.** Grid and proposal boxes are merged (IoU $> 0.9$ deduplication, grid first). If fewer than 8 remain, deterministic fallback boxes are appended: $(0.35, 0.35, 0.65, 0.65)$, $(0.05, 0.05, 0.35, 0.35)$, $(0.65, 0.65, 0.95, 0.95)$ — deliberately smaller boxes so a fallback probes a different scale rather than duplicating the grid. If still short (heavy overlap), centered square boxes of side $0.45, 0.40, 0.35, \ldots$ (step 0.05, floor 0.12) are added. The candidate set is thus a deterministic function of the sample and the single proposal generation.

**Crop semantics.** Crops are taken from the *original* native-resolution image, then re-encoded at the crop budget; cropping the coarse view would make CROP an upsampling artifact rather than an information-acquiring action. A crop view is *added* to the coarse overview (global context is retained), with a view label naming its normalized coordinates.

**Deployable Crop-family semantics.** For the deployed one-step selectors, CROP-FAMILY does not denote outcome-dependent selection of the best crop. Its utility and cost are the uniform expectation over the fixed eight-crop family. Operationally, selecting CROP-FAMILY corresponds to sampling one of the eight fixed crops uniformly at deployment. Because all eight crop outcomes are executed in the offline audit, the reported family-level outcome and visual-token cost are the exact expectation of this realizable randomized policy. The unrestricted structural oracle may instead select among individual crop outcomes and is therefore a distinct counterfactual benchmark (Section 9.13).

### 2.2 Control (dummy) actions and the luck floor

To price in "the reader was simply asked again," the audit includes a dummy arm of *matched cardinality* (11 actions, same as the real action set):

- **Size-matched random crops**: same count and same width/height distribution as the real candidates, position drawn uniformly from a seed derived deterministically from the sample id (`ppu-dummy::{sample_id}`). Because area and aspect ratio are matched, "question-conditioned beats random" cannot be a crop-size effect in disguise.
- **Re-encode dummies**: the coarse view re-rendered at the same budget with a different resampling filter (Lanczos, or bilinear plus a JPEG-90 round trip), i.e. cosmetically different pixels with no new information. Two distinct variants exist so the dummy pool holds 11 *distinct* draws.

The dummy oracle serves as a diagnostic control for selection-on-noise (the "luck floor" of the structural audit) rather than as the comparator defining the reported cross-replay headroom $H_{\mathrm{XR}}$, which is the cross-replay oracle minus the cross-fitted state-independent best fixed arm. For question-conditioned localization specifically, size-matched random boxes with identical area and aspect ratio (position-only randomization) are the comparison class.

### 2.3 OCR as an external tool

- Engine: RapidOCR (ONNX runtime, CPU). The OCR engine is deliberately *not* the VLM: if the reader transcribed the image itself, OCR and HighRes would be the same action and action-diversity results would be a prompt-wording artifact.
- OCR reads the *original* native-resolution image, not the coarse view — extracting text the reader cannot resolve is the point of the tool.
- Serialization: detected lines are sorted into reading order (by rounded $y_0$, then $x_0$), truncated to 30 lines, and rendered one line per row as `[x0,y0,x1,y1] text` with normalized coordinates at two decimals. An empty result is rendered as `(no text detected)`.
- Cost accounting: engine latency and the tool call are recorded as cost. Campaigns use a pre-pass cache (each image OCRed once across CPU workers), but the *recorded* latency is the engine's real cost, not the cache lookup's, so the cost column reports what a deployed agent would pay. OCR output is an observation, produced only after the action is chosen; nothing upstream of the action choice may read it.

### 2.4 HighRes and Stop

HighRes replaces the coarse overview with a higher-resolution full view at the dataset's HighRes budget. Stop answers from the coarse view alone. In the sequential experiment, HighRes as a second step replaces the coarse overview while *retaining* evidence already acquired by the first action.

## 3 Prompts (Verbatim)

One template serves every arm; the only thing that varies across actions is the observation block. If actions had their own instructions, action comparisons would partly measure prompt engineering. The prompt version is logged with every rollout and is part of the generation-cache key.

**System prompt.**

```
You are answering questions about images. You may be shown a low-resolution overview, additional
    zoomed-in views, or text extracted from the image. Use only what you are shown. Answer as
    briefly as possible.
```

**Open-ended instruction, original (`p0.1`).**

```
Answer the question with a short phrase (a few words at most). End your reply with a line of the
    form:
Answer: <your answer>
```

**Open-ended instruction, one-word control (`p0.2-oneword`).** Used only in the answer-format intervention arms:

```
Answer with a single word or number. For yes/no questions answer exactly 'yes' or 'no'. Do not
    write a sentence. End your reply with a line of the form:
Answer: <your answer>
```

**Multiple-choice instruction.**

```
Answer with the letter of the correct option. End your reply with a line of the form:
Answer: <letter>
```

**User-turn assembly.** The user turn concatenates, in order and separated by blank lines: (i) a view inventory, e.g. "`Views provided, in order: coarse@192; zoomed view of region 0.45,0.00,1.00,0.55.`"; (ii) if OCR was invoked, "`Text extracted from the image by an OCR tool:`" followed by the serialized block of Section 2.3; (iii) the question as served under the current replay (options inlined as `A.`–`E.` lines for multiple choice); (iv) the applicable instruction above. View labels describe what an observation *is* (coordinates are part of the action specification), never the content of an observation the agent has not paid for.

**Crop-proposal prompt.**

```
Question: {question}

Look at the image and the question. Name up to 3 regions you would need to inspect at higher
    resolution to answer it. Reply with one region per line, each as four numbers between 0 and 1:
    x0,y0,x1,y1 (left, top, right, bottom as fractions of the image). No other text.
```

# 4 Replay Construction

Decoding is greedy, so re-running a request returns the same string; a replay therefore cannot and does not measure decoding noise. What threatens the audit is *luck* — an action that happens to push a 5-way multiple choice onto the right letter, or a phrasing the reader latches onto. R2 changes what is asked, never how it is sampled. Both replays are deterministic functions of the sample id and are reproducible from the manifest alone.

**Multiple choice (MME-RealWorld).** R2 permutes the answer options with a seeded RNG (`ppu-r2::{sample_id}::7`) and follows the gold answer to its new letter. If the drawn permutation is the identity, the order is reversed instead — a permutation that changes nothing is not a control. Scoring under R2 uses the moved gold letter.

**Open-ended (TextVQA, VQAv2).** R2 applies a fixed, meaning-preserving, syntax-level rewrite — no LLM sampling. Rules are applied in order, first match wins (case-insensitive, anchored at both ends):

| Pattern | Rewrite |
|---|---|
| `what is …?` | `Looking at the image, what is …?` |
| `what are …?` | `Looking at the image, what are …?` |
| `what …?` | `Based on the image, what …?` |
| `who/where/when/which/how …?` | `Based on the image, wh-word …?` |
| `is/are/does/do …?` | `Based on the image, is/are/does/do …?` |
| (no match) | `Based on the image: …` |

The rewrite set is deliberately small and mechanical: a paraphrase that changed difficulty would surface as fake instability in the label-reliability numbers. Multi-replay robustness checks (replays 3–5, prefix variants) are summarized in Section 9.6.

# 5 Scoring Rules

**Answer extraction.** The answer span is pulled from the raw generation in a fixed, deliberately forgiving order: (1) an `<answer>...</answer>` tag; (2) a line matching `(final )?answer[:-]`; (3) the last non-empty line. A parser that silently preferred one surface form would bias whichever action makes the model chatty. For multiple choice, the first standalone letter A–E in the extracted span (falling back to the full text) is used. `parse_ok=False` (no answer found) scores 0 but is tracked separately: a rising parse-failure rate indicates a pipeline bug, not a model result.

**Strict scorers (primary).** No LLM judge anywhere; every metric is deterministic and recomputable offline from the stored generations.

- `vqa_acc` (TextVQA, VQAv2): official VQA accuracy. Normalization follows the official evaluator (lowercasing, punctuation stripping with number-comma and decimal-period handling, contraction restoration, word-to-digit mapping for zero–ten, article removal); the score is $\min(m/3, 1)$, where $m$ is the number of exact matches among the 10 references.
- `mcq` (MME-RealWorld): extracted letter vs. gold letter under the current replay's option order (first character, case-insensitive). Because scoring is letter-level, lenient and strict scoring coincide on MME by construction.
- Binary quantities (HELP/HARM/DMG, strict labels) threshold soft scores at 0.5.

**Format-tolerant (lenient) rescoring (secondary).** As defined in the main paper: for normalized prediction $p$ and reference $g_j$, a reference counts as matched if $g_j = p$, or $g_j$ equals the first word of $p$, or $g_j$ occurs in $p$ as a whitespace-bounded substring; the score is $\min(\sum_j h(g_j, p)/3, 1)$. Lenient evaluation changes only the mapping from an existing generation to its score — generations, observations, and replays are identical bytes.

**Confidence.** Primary confidence is the mean selected-token log-probability over the complete STOP reply, averaged across the two STOP replays. Answer-span mean, span-minimum, and first-token variants are computed from the already-logged per-token probabilities (Section 9.4).

# 6 Diagnostic and Selector Features

The frozen v2.2 feature set (also "F0" in the selector experiments) contains 16 inexpensive pre-action features (`ppu/gate_features.py`). By design it contains *no* standalone confidence term: an earlier version learned a single global direction ("confident, therefore look") that fit VQAv2 — where confidence is inverted under the original prompt — and destroyed TextVQA. Confidence therefore enters only through interactions, so the model can decide per answer form whether the uncertainty is worth believing. Deliberately absent: dataset id, image resolution, native token count — features that name the domain rather than the state.

| Group | Feature | Definition |
|---|---|---|
| Answer surface | `ans_chars` | characters in the provisional STOP answer |
| | `ans_tokens` | generated tokens in the answer |
| | `ans_words` | whitespace words in the answer |
| | `is_numeric` | answer is a bare number |
| | `is_yesno` | answer starts with yes/no |
| | `has_hedge` | hedge regex[1] fires on the answer |
| Question surface | `q_chars`, `q_words` | question length |
| | `q_is_howmany` | contains "how many" |
| | `q_is_what` | starts with "what" |
| | `q_is_isare` | starts with is/are/does/do/was/were |
| Confidence (interaction only) | `conf_x_yesno` | conf_rank × `is_yesno` |
| | `conf_x_numeric` | conf_rank × `is_numeric` |
| | `conf_x_other` | conf_rank × `is_other` |
| | `conf_x_len` | conf_rank × min(`ans_chars`/20, 2) |
| | `conf_x_hedge` | conf_rank × `has_hedge` |

conf_rank $\in [0, 1]$ is the empirical CDF of the STOP mean-token log-probability, fitted on training states only (0 = most confident). Features are standardized with training-set statistics. The binary v2.2 diagnostic uses class-balanced logistic regression on these features; the multi-action selector uses per-family ridge value heads on the same features (Section 7).

# 7 Frozen Policies, Baselines, and Artifacts

Each learned component is trained only on its designated training pool and frozen before outcomes from its own prospective evaluation set are observed. Freeze artifacts are stored (pickled) with JSON manifests recording the relevant weights, scaler statistics, confidence ECDF, feature list, and seed. Freeze artifacts and their SHA-256 prefixes are included in the ZIP (`models/`).

**One-step selector v2 (primary prospective test).** Chosen by five-fold development cross-validation from a pre-specified 10-candidate grid (2 feature sets × 5 head classes), selection criterion TextVQA policy value at 1/2× budget, tie-break 1/4×. Winner: F0 features + ridge regression ($\alpha = 0.3$). Three heads predict the replay-averaged score increment of each family (CROP, OCR, HIGHRES) relative to STOP; the CROP target and cost use the deployable uniform-grid family expectation (no outcome-dependent region selection). Decision rule: $\arg\max_f [\widehat{\Delta}_i(f) - \lambda_B c_i(f)]$ with zero predicted gain and cost for STOP; $\lambda_B$ selected on development data inside the frozen artifact's $\lambda$ grid, per budget. Artifact: `selector_frozen_v2` (sha256[:16] `d4e495063ac56b59`).

**State-independent randomized mixtures.** For each budget, an LP over the fixed arms {STOP, each grid/fallback crop box, uniform CROP, OCR, HIGHRES} maximizes the mean fixed-arm outcome subject to one expected-token constraint and a simplex constraint; with one linear constraint the optimum mixes at most two arms (concave envelope). Three comparator strengths are reported: in-sample (weights fitted on the evaluation sample — a baseline-favoring upper bound), cross-fitted (2-fold), and deployable (fitted on development data, applied unchanged). The frozen deployable TextVQA mixture is: 1/4×: $0.876 \cdot$ CROP-center $+ 0.124 \cdot$ OCR; 1/2×: $0.569 \cdot$ CROP-center $+ 0.431 \cdot$ CROP-top-right; 1×: $1.0 \cdot$ CROP-top-right.

**Hidden-state selector v3.** Six candidate configurations compared on development data before any fresh outcome was observed; frozen winner: the reader's STOP-time hidden representation with ridge value heads ($\alpha = 30$), same value-minus-cost decision rule. Artifact: `selector_frozen_v3` (sha256[:16] `4fd1fa4f`). Development diagnostics recorded before prospective collection flagged both weaker development performance than F0 and weak $\lambda$-budget controllability; both materialized prospectively (Section 9.10).

[1] `maybe|possibly|probably|appears|seems|likely|unclear| cannot|can't|unable|not sure|hard to|difficult|blurry|unknown|unreadable| too small` (word-bounded, case-insensitive).

**F0 nonlinear-head prospective follow-up.** A separately preregistered follow-up (`PREREG_STRONG_CONTROLLER`, 09-20) tests whether the prospective gap is specific to the ridge value head. Ridge, XGBoost, and MLP heads use the same F0 decision-time representation, utility targets, costs, and value-minus-cost decision rule. Training uses a frozen 1,350-state pool assembled from previously analyzed development and evaluation states (Gate-0 200, held-out 250, prospective 600, v3 300); no state from the new 400-state evaluation set is included. Per head class one configuration is chosen from a disclosed grid by nested five-fold cross-validated policy value at 1/2×; $\lambda_B$ per class is chosen on pooled out-of-fold predictions. The preregistered primary comparison is F0/XGBoost minus the frozen state-independent mixture at the strict 1/2× operating point; other heads, budgets, and format-tolerant analyses are secondary. Artifact: `strong_controllers_frozen` (sha256[:16] `5c60ccab3f7d08b8`).

**Representation–head $3 \times 3$ prospective test.** The final prospective controller test (`PREREG_HIDDEN_CONTROLLER`, 09-20) crosses three decision-time representations—F0, the 3,584-dimensional STOP-time hidden representation H, and H+F0—with ridge, XGBoost, and MLP value heads. All nine configurations use the same 1,750-state training pool (the 1,350 states above plus the 400-state nonlinear-head evaluation set) and the same family-level value-minus-cost decision rule; parameters and per-budget $\lambda_B$ are frozen before evaluation on a new 400-state TextVQA set, and the primary nonlinear cell (H/MLP, by the same cross-validation rule) is recorded in the frozen manifest before collection. The strict 1/2× H/MLP-minus-mixture comparison is preregistered as primary; other representation–head cells, budgets, and format-tolerant analyses are secondary. Artifact: `hidden_controllers_frozen` (sha256[:16] `ed93f19437264e56`).

**Sequential controller v1 (original, preregistered test).** Two post-first-observation features: whether the normalized first-step answer differs from the original STOP answer, and first-step answer length. Two ridge heads ($\alpha = 1.0$) predict the incremental value of $\text{CROP}_{\text{TR}}$ and HIGHRES over stopping; decision $\arg\max(0, \widehat{\Delta}_{\text{TR}}, \widehat{\Delta}_{\text{HI}})$; no cost term. Strict-trained and lenient-trained variants both frozen (`seq_controller_frozen`, sha256[:16] `cffa1043f455f7d1`). The frozen state-independent continuation mixture, fitted on the 100-state development audit, places all mass on $\text{CROP}_{\text{TR}}$ (i.e., always-$\text{CROP}_{\text{TR}}$).

**Sequential controller v2 (disclosed cost-aware follow-up).** Seven history features (including confidence before/after the first observation and their change), ridge continuation-value heads, and explicit cost: $\arg\max_a[\widehat{\Delta}(a \mid h_1) - \lambda_B c(a)]$. Trained and frozen on the same 100 development states (`seq_controller_v2_frozen`, sha256[:16] `b74b8b6e70332a52`); evaluated on the already-separated 300-state set as a disclosed secondary analysis.

**Budget-constrained oracles.** The Lagrangian sweep uses $a_i(\lambda) = \arg\max_a[y_i^{(\text{sel})}(a) - \lambda c_i(a)]$ over a finite $\lambda$ grid ($\lambda_{\text{tok}} \in \{0\} \cup \text{logspace}(-6, -2, 33)$; $\lambda_{\text{lat}} \in \{0\} \cup \text{logspace}(-5, -0.5, 33)$), selected on one replay and scored on the other. The exact multiple-choice-knapsack DP closes the relaxation loophole (Section 9.2). Both are counterfactual and non-deployable by construction.

# 8 Preregistration Index and Deviations Log

All preregistration documents are included verbatim in `preregistrations/`. Each was written and frozen before the corresponding outcomes existed; "post-hoc relative to the paper, but preregistered relative to its own data" is stated explicitly where it applies.

**Preregistration chronology.** Every row lists the date the document was frozen, the SHA-256 prefix of the frozen artifact(s) and evaluation manifest recorded *in* the document, the collection window from the run metadata of the log, the single preregistered primary endpoint, and the outcome as reported. "Pool" rows train on sets that had already been analyzed (disclosed in the document); evaluation sets are never reused.

| Document | Frozen | Frozen artifact (sha256[:16]) | Evaluation set ($n$; sha) | Collected | Primary endpoint | Outcome |
|---|---|---|---|---|---|---|
| FRESH_LOCKBOX | 08-23 | frozen confidence + v2.2 gate (unchanged) | VQAv2 250 + TextVQA 250 fresh train states | 08-23 | Format-confound replication on fresh states | Replicated (Sec. 9) |
| MULTIACTION_ HELDOUT | 08-28 | multi-action policy v1 (dev-fitted) | TextVQA 250 + VQAv2 250 held-out | 08-28–30 | Multi-action − best fixed at 1/4×, 1/2× (TextVQA, strict) | PASS: +7.0 [+2.7, +11.4] / +9.9 [+5.5, +14.4] |
| SECOND_READER | 08-30 | protocol only (Phi-4 reader) | Gate-0 states, 196 MME + 200 TextVQA | 08-30 | Pooled symmetric headroom > 0 (H1) | PASS: pooled +6.4; TextVQA +10.0 [+6.0, +14.0] |
| MULTI_REPLAY | 08-30 | protocol only (replays 3–5) | 100-state subsets, both datasets | 08-30 | Protocol validation, no threshold | $K$=2 sufficient; $K$=2 estimate conservative |
| HUMAN_AUDIT | 08-28 | blinded sheets (never opened) | 381 + 489 items | — | Strict vs. human label transition | **Withdrawn 08-30; declined again 09-01 and 09-20** |
| HUMAN_ ADJUDICATION | 09-21 | blinded 300-item package 1384f99672bc8d65 | prospective-600 generations: 200 scorer disagreements + 100 controls | 09-21 (3 annotators) | $P$(human correct \| strict = 0, tolerant > 0) | Executed; protocol frozen before labels: **.807 [.746, .856]**; Fleiss' $\kappa$ .766 |
| PROSPECTIVE_ SELECTOR | 08-31 | selector v2 d4e495063ac56b59; mixture (dev-fitted) | TextVQA 600; a535d8edce3aa822 | 08-31 | v2 − mixture, 1/2× strict | **primary non-confirmatory**: +0.9 [−1.6, +3.4]; lenient +4.6 [+2.4, +6.8] |
| SEQUENTIAL | 08-31 | protocol only (two-step audit) | TextVQA 100 | 08-31 | Sequential oracle − stop-after-$a_1$ | +9.3 [+4.7, +14.5] (center) |
| SEQ_PROSPECTIVE | 08-31 | seq. controller v1 cffa1043f455f7d1 | TextVQA 300; 4196e1abbead1506 | 08-31 | Controller − always-CROP$_{\text{TR}}$, center, strict | **primary non-confirmatory**: −1.7 [−3.6, +0.2] |
| SELECTOR_V3 | 09-01 | selector v3 (H + ridge) 4fd1fa4fc1793b7a | TextVQA 300; 7bc5ec8a8eb9a61f | 09-01 | v3 − mixture, 1/2× strict | **primary non-confirmatory**: +1.7 [−1.9, +5.1]; $\lambda$-budget uncontrollable |
| STRONG_ CONTROLLER | 09-20 | F0 × {ridge, xgb, mlp} 5c60ccab3f7d08b8 (pool 1,350) | TextVQA 400; 7312457434be55f3 | 09-20 | xgb − mixture, 1/2× strict | **primary non-confirmatory**: +1.3 [−1.4, +4.1]; xgb − ridge −0.3 [−2.0, +1.6] |
| HIDDEN_ CONTROLLER | 09-20 | {F0, H, H+F0} × {ridge, xgb, mlp} ed93f19437264e56 (pool 1,750; primary cell H/mlp) | TextVQA 400; af41a4ac740a9ecb | 09-21 | H/mlp − mixture, 1/2× strict | **primary non-confirmatory**: +0.8 [−2.1, +3.8]; H/xgb − F0/xgb −3.1 [−5.8, −0.6] |

Zero-inference analyses added in the 09-20/21 round (no preregistration needed; existing logs only): cross-reader agreement conditional on baseline capability, replay-pairwise robustness (all ordered replay pairs, and the R1/R2/R3 triangle at $n$=400), and the action-matched restricted oracle (Section 9.13).

**Original human audit withdrawal and later adjudication.** The original blinded semantic-utility audit preregistered on 08-28 was withdrawn on 08-30 before any human verdict was entered; its sheets remained empty and no result from that protocol exists.

A distinct human-adjudication study was later motivated post hoc, after the automatic scorer-sensitivity results were observed; its sampling and annotation protocol (PREREG_HUMAN_ADJUDICATION, 09-21) were preregistered before any human label was collected. This later study sampled 300 unique (state, normalized answer) pairs from the 600-state prospective TextVQA pool: 200 benchmark-negative, format-tolerant-positive disagreement pairs and 100 agreement controls. Annotators were blinded to scorer outputs, action arm, replay, and policy identity. This later adjudication is treated throughout the paper as retrospective validation rather than preregistered primary evidence.

**Deviations and incidents (complete log).**

1. **Engine death during held-out collection (08-28).** A vLLM engine crash poisoned 183 VQAv2 held-out states with transport-error rollouts (the collector logged errors instead of aborting). The

affected rollouts were stripped (backups retained) and re-collected *before* any evaluation was run. All subsequent drivers carry a zero-transport-error integrity gate; every reported collection passed it.

2. **scikit-learn version shim.** Frozen sklearn-1.9.0 pickles were evaluated under 1.7.2 via a `predict_proba` attribute shim, asserted equal to the sigmoid of the frozen coefficients to $10^{-16}$.
3. **Human audit withdrawn without execution.** The blinded human semantic-utility audit (preregistered 08-28) was withdrawn on 08-30 before any verdict was entered; the blinded sheets remain empty and the private mapping unopened. No result exists to report or suppress. The paper's "measurement sensitivity" framing (rather than a claim about which scorer is semantically correct) reflects this: a separately proposed adjudication protocol was declined twice (09-01, 09-20) and then executed on 09-21 under a third, narrower preregistration (`HUMAN_ADJUDICATION`; Section 9.14).
4. **Manifest regeneration incident (09-01).** `textvqa_prospective.jsonl` was accidentally regenerated with a changed exclusion pool by an old builder script; it was restored immediately by reconstructing the original 950-id exclusion set. The restored file matches the preregistered SHA-256 prefix `a535d8edce3aa822` exactly and its ids match the collected log 600/600. No analysis touched the corrupted file.
5. **Phi-4 weights re-download.** The HuggingFace cache had been purged between sessions; weights were re-downloaded and the serving configuration reproduced from the original serving log.
6. **Hidden-controller pipeline relaunches (09-20/21).** The `HIDDEN_CONTROLLER` pipeline was launched three times: the first attempt failed before any inference because the held-out set is a train-split draw and the extraction script was pointed at the val manifest (fixed); the second and third attempts lost the shared GPU to other users' jobs during the 1.5-hour training step and at server start. The frozen artifact (`ed93f19437264e56`, 09-20 15:59) was never retrained, no evaluation state was collected before the final launch (09-21 09:56), and the fresh-set hidden states were extracted after collection from STOP-time inputs only. The training pool intentionally includes every previously analyzed TextVQA set (1,750 states); the evaluation set is disjoint from all of them.
7. **Adjudication file drop (09-21).** The first set of verdict files contained a duplicated file (annotator 1 and annotator 3 byte-identical). The preregistered single-pass analysis was not run on it; the correct third file was supplied and the analysis ran once on three distinct files (MD5 prefixes `d1ced72c`, `24f29c79`, `82d06afc`).
8. **Strong-controller collection (09-20).** Single launch; 400/400 states, 13,200 rollouts, zero transport errors; the same log carries replay 3 for the replay-robustness extension.
9. **Sequential controller v2 confidence regeneration.** Development first-step confidence required regeneration; byte-match was 379/400 against the original generations (disclosed in the result file).
10. **Comparator strengthening (disclosed post-hoc).** The original structural comparator was the best deterministic fixed arm; randomized state-independent mixtures were added as a disclosed post-hoc strengthening, and all mixture-based conclusions are labeled accordingly.

# 9 Additional Results

All confidence intervals resample independent perceptual states (10,000 bootstrap draws); policy differences use state-paired resampling. "pp" denotes percentage points. Unless noted, scoring is strict.

## 9.1 Budget-constrained oracle frontier (Gate-0 logs, Qwen)

Cross-replay symmetrized, deterministic arms; budget expressed as a fraction of the always-HighRes additional-token cost.

| Budget | MME Δ vs. Stop (recovery) | TextVQA Δ (recovery) |
|---|---|---|
| 5% | +.122 (96%) | +.112 (59%) |
| 10% | +.128 (100%) | +.184 (97%) |
| 20% | +.128 (100%) | +.190 (100%) |

## 9.2 Exact constrained oracle vs. Lagrangian relaxation

At ≥10% budget the Lagrangian read-off matches the randomized hull to ≤1pp. At the 5% budget the exact multiple-choice-knapsack DP is *higher* by +1.5pp (MME) and +2.8pp (TextVQA): closing the relaxation loophole strengthens the front-loading conclusion. The DP's cross-replay evaluated value can sit ∼1pp below the hull at loose budgets because exact optimization of the selection replay slightly overfits selection noise — itself a small illustration of selection inflation.

## 9.3 Per-action damage decomposition

$P(\text{harm} \mid \textsc{Stop}\ \text{correct})$, existential over the family ("any") vs. single actions, Qwen:

| Family | MME any / single | TextVQA any / single |
|---|---|---|
| Crop (grid) | .449 / .169 | .154 / .049 |
| OCR | .080 / .080 | .085 / .085 |
| HighRes | .145 / .145 | .043 / .043 |
| Any action (DMG) | .464 / .157 | .222 / .051 |

Aggregate DMG (13.8/13.0%) is dominated by the grid-crop existential; single-action harm is 4–17%. On TextVQA, $P(\text{help} \mid \textsc{Stop}\ \text{wrong}) = .782 \gg P(\text{harm} \mid \textsc{Stop}\ \text{correct}) = .222$. Phi-4 shows the same structure with lower DMG (7.7/7.0%) but higher OCR harm on MME (40% vs. Qwen's 10.7%).

## 9.4 Answer-span confidence variants

Regeneration byte-match 100%. TextVQA ROC–AUC is stable at 0.74–0.80 under all span variants. VQAv2 under the original prompt: reply-mean 0.42–0.43 (below chance), recovering only to 0.58–0.62 at span-minimum; under the one-word prompt 0.72–0.90. The confidence pathology is a generation-format phenomenon, not a token-aggregation artifact.

## 9.5 Second reader (Phi-4): lenient rescoring and realizable signal

TextVQA lenient rescoring of the same Phi-4 generations: Stop .258 → .505 (format artifact confirmed); cross-fitted best fixed .624; cross-replay oracle .743; headroom +11.9pp [+7.1, +16.7]; Stop-optimal 36.9%; frontier recovery 61% at 5% budget, 90% at 10%. MME is multiple-choice, so lenient = strict by construction. Phi-4's Stop verbosity under this protocol (4.0 words mean, 32% >3 words, vs. 1.5 / 5% in its own native-format runs) explains its low absolute strict accuracy; structural quantities, not absolute levels, are the comparanda.

Phi-4's own Stop-confidence selector is indistinguishable from random allocation at every matched budget (confidence − random point estimates −0.003 to +0.013, all CIs crossing 0), while the constrained oracle stands far above both (TextVQA 10% budget: random .251, confidence .254, oracle .396). Headroom replicates across readers; the zero-training realizable signal does not.

## 9.6 Multi-replay robustness (replays 3–5)

On 100-state outcome-blind subsets: the naive same-replay oracle stays inflated at every $K$ (MME .65–.66 vs. cross-replay .50–.53; TextVQA .885–.888 vs. .773). $K$=4–5 estimates stay inside the $K$=2 bootstrap CI on both datasets (pre-declared sufficiency criterion). TextVQA held-out mean and symmetric estimates *rise* with $K$ (.773 → .830/.836): more evaluation replays denoise the outcome, so the published $K$=2 headroom is conservative. Headroom is positive at every $K$.

**All ordered replay pairs, and a three-replay follow-up on 400 states.** On the same 100-state subsets every ordered replay pair ($r \to s$) gives a positive cross-replay headroom (TextVQA 20/20 pairs, mean +.110, sd .035, 16 of 20 CIs excluding zero; MME 20/20, mean +.070), and the published R1→R2 cell is the smallest of the 20 on the TextVQA subset. On the later 400-state nonlinear-head follow-up,

an independently constructed third replay ("In the image shown: …") permits all six ordered directions among replays 1–3: headroom R1→R2 +.095 [+.055, +.135], R1→R3 +.103 [+.068, +.138], R2→R3 +.113 [+.082, +.147] (reverse directions +.095 / +.102 / +.102); STOP agreement ranges from .945 to .980, per-arm action-score correlation from .897 to .933, and sign agreement of the per-action utility among both-nonzero pairs from .97 to 1.00. Files: `replay_pairwise_robustness`, `replay_pairwise_strong`.

## 9.7 Randomized-mixture baselines on the held-out sets

TextVQA strict, selector − mixture (paired bootstrap):

| Budget | vs. in-sample | vs. cross-fitted | vs. deployable |
|---|---|---|---|
| 1/4× | +.009 [−.027, +.045] | +.020 [−.017, +.056] | +.010 [−.025, +.046] |
| 1/2× | +.012 [−.022, +.046] | +.018 [−.012, +.049] | +.030 [−.003, +.063] |
| 1× | −.011 [−.045, +.024] | −.011 [−.045, +.024] | +.028 [−.009, +.065] |

VQAv2 strict vs. cross-fitted: −.153 [−.204, −.102] at 1/4× and −.072 [−.117, −.027] at 1/2× (a fixed grid crop significantly beats the selector); under lenient rescoring the reversal vanishes (+.010 [−.017, +.038], +.014 [−.017, +.045]). The deployable mixture transferred from development to held-out with no loss (.717 development fit → .744 held-out at 1/2×), so a transfer-loss explanation is not available.

## 9.8 Prospective one-step evaluation: realized-cost matching

Re-solving the development mixture at the selector's *realized* token cost (53/129/212) on the 600 prospective states: strict +1.9 [−0.1, +4.0] / +1.2 [−1.4, +3.8] / +0.9 [−1.7, +3.4] (all non-confirmatory); lenient +2.7 [+0.9, +4.5] / +5.1 [+2.8, +7.4] / +3.6 [+1.4, +5.9] (all excluding zero). Cost-matching therefore strengthens, not weakens, the scorer-dependence finding.

## 9.9 Sequential experiments: full cell table

Development audit ($n$=100): cross-replay oracle over {stay, → $\text{CROP}_{\text{TR}}$, →HIGHRES} beats stopping after the first action by +9.3pp [+4.7, +14.5] (center-crop branch) and +9.2pp [+4.2, +14.5] (OCR branch). Continuation gain concentrates on first-step failures (+22 to +27pp) and is ≈0 on successes. "First-step answer changed" predicts residual utility with AUC .70/.74 (base rate of "second look helps" ≈ 10–15%); answer length alone ≈ .55. At matched tokens the best one-step arm (CROP-TR, .767 at 243 tokens) matches or beats every fixed two-step chain: the sequential value is adaptive, not a fixed pipeline.

Prospective test ($n$=300), controller − frozen continuation (always-$\text{CROP}_{\text{TR}}$):

| Branch | Scorer | v1 (original) | v2 (cost-aware, disclosed) |
|---|---|---|---|
| Center | strict | **−1.7** [**−3.6**, **+0.2**] (primary) | −0.6 [−1.8, +0.6] |
| Center | lenient | −1.2 [−2.9, +0.4] | — |
| OCR | strict | −0.3 (ns) | +0.4 (ns) |
| OCR | lenient | +2.7 [+0.4, +5.1] | +2.7 [+0.5, +4.8] |

Both v1 variants collapse to look-rate 1.00 on fresh states; v2 restores stopping (look-rate 0.83–0.85, or 0.30 under the conservative $\lambda_B$, which matches always-$\text{CROP}_{\text{TR}}$ within −1.6pp ns at 80 vs. 241 tokens — one third the cost). Sequential oracle headroom on the same 300 states (all eight CIs exclude zero): oracle − continuation = +2.8 [+1.1, +4.8] (center strict), +5.8 [+3.4, +8.4] (OCR strict), +2.7 [+1.0, +4.7] / +6.2 [+3.8, +8.8] lenient; oracle − stop-after-first = +4.4 [+1.9, +6.9] / +14.0 [+10.3, +17.8]. Only one of eight secondary controller cells is positive with a CI excluding zero (multiplicity caveat in the main paper).

## 9.10 Hidden-state selector v3: budget-control failure

Primary (1/2× strict, v3 − mixture): +1.7pp [−1.9, +5.1]; 1/4×: +2.9 [−0.7, +6.3]; 1×: +1.7 [−2.0, +5.6]. The pre-flagged instability materialized exactly: realized cost is ≈174–179 tokens at *every* nominal budget

— $\lambda(B)$ does not bind with hidden-state value heads. The 1/4× and 1/2× point estimates are therefore cost-confounded (2.3× / 1.1× the mixture's tokens) and are not claimed. The only cost-fair cell is 1×, where v3 is cheaper than the mixture (179 vs. 241 tokens) at +1.7pp ns strict / +3.5 $[-0.0, +7.2]$ borderline lenient, and v3 ≈ v2 (+0.2pp ns). Secondary v3 − v2 at 1/4× (+4.5 $[+1.0, +8.1]$) is likewise cost-confounded (174 vs. 53 tokens) and not claimed as a win. Beyond the non-confirmatory primary, this surfaces a deployability failure mode — loss of $\lambda$-budget controllability — that the handcrafted selector does not have.

### 9.11 F0 nonlinear-head prospective follow-up

On 400 fresh TextVQA states (manifest sha256[:16] `7312457434be55f3`; 13,200 rollouts including a third replay, zero transport errors) at the strict 1/2× operating point, F0/XGBoost reaches .726 at 148 visual tokens, F0/ridge .728 at 141, and F0/MLP .716 at 160, compared with .713 for the frozen state-independent mixture, .777 for the action-matched oracle, and .812 for the unrestricted oracle. The preregistered XGBoost-minus-mixture primary comparison is +1.3 points (95% CI $[-1.4, +4.1]$) and therefore non-confirmatory. Capacity contrasts are null at every budget (XGBoost − ridge −0.3 $[-2.0, +1.6]$, MLP − ridge −1.3 $[-3.2, +0.6]$ at 1/2×); the pool-trained ridge exceeds the published dev-only v2 by about 3 points (.728 vs. .699). Realized costs track the nominal budgets for all three heads (54–78 / 141–160 / 210–225 tokens). Secondary format-tolerant XGBoost-minus-mixture differences are positive at 1/2× (+3.3 $[+0.7, +6.1]$) and 1× (+4.4 $[+1.4, +7.5]$), with XGBoost − ridge still null. These results reduce the concern that the original prospective result is specific to a linear ridge head, but they do not bound richer controller classes. File: `strong_controller_eval`.

### 9.12 Representation–head $3 \times 3$ prospective test

On a further 400 fresh TextVQA states (manifest sha256[:16] `af41a4ac740a9ecb`; 8,800 rollouts, zero transport errors; hidden states extracted after collection from STOP-time inputs only), at the strict 1/2× operating point, accuracies (realized visual-token cost) are:

| | F0 | H | H+F0 |
|---|---|---|---|
| Ridge | .749 (146) | .729 (163) | .732 (161) |
| XGBoost | .737 (141) | .705 (151) | .724 (131) |
| MLP | .744 (122) | .722 (171) | .707 (142) |

The mixture reaches .714, the action-matched oracle .777, and the unrestricted oracle .808. The preregistered H/MLP-minus-mixture primary comparison is +0.8 points (95% CI $[-2.1, +3.8]$) and non-confirmatory; H/XGBoost − mixture is −0.8 $[-3.6, +2.0]$. H/XGBoost underperforms F0/XGBoost by −3.1 points ($[-5.8, -0.6]$; −5.9 at 1/4×, −2.5 at 1×) and H/ridge by −2.4 ($[-4.8, -0.1]$); H/ridge − F0/ridge is −1.9 $[-4.2, +0.4]$ and H+F0/XGBoost − F0/XGBoost −1.3 $[-3.3, +0.7]$. Realized costs track the nominal budgets for all nine cells (41–73 / 122–171 / 162–237 tokens), so the v3 budget-control failure does not recur with pooled out-of-fold $\lambda$. A secondary, unadjusted F0/ridge comparison is +3.5 points ($[+0.8, +6.4]$) versus the mixture and is treated as hypothesis-generating (the same contrast was +1.6 $[-1.2, +4.4]$ on the nonlinear-head set). Under format-tolerant rescoring, all nine configurations exceed the mixture at 1/2× by +2.8 to +5.2 points with unadjusted confidence intervals excluding zero, and the H − F0 contrasts are null; these secondary analyses are not part of the preregistered strict primary test. Pre-collection nested cross-validation on the training pool had already ranked every H-based cell below F0 (recorded in the frozen manifest). File: `hidden_controller_eval`.

### 9.13 Action-matched counterfactual oracle

The unrestricted structural oracle may choose among individual crop slots {STOP, $\text{CROP}_{1..8}$, OCR, HIGHRES}, whereas the deployed selectors choose only among {STOP, CROP-FAMILY, OCR, HIGHRES}. To isolate the contribution of this action-space mismatch, we recompute the cross-replay oracle over the deployed family-level action space, using the uniform eight-crop expectation for CROP-FAMILY (outcome and cost exactly as the selectors use them), under the audit protocol (select on replay 1, ties to the cheaper family,

score on replay 2) and against the *same* cross-fitted best fixed arm, so that $H^{\text{full}} - H^{\text{dep}}$ isolates the action space. Both oracles remain counterfactual because their action choice uses replay outcomes. This analysis is post hoc (zero inference on the existing logs) and is not a preregistered primary endpoint. Strict:

| Set | $n$ | $H^{\text{full}}$ [CI] | $H^{\text{dep}}$ [CI] | matched/unrestricted |
|---|---|---|---|---|
| TextVQA Gate-0 | 200 | $+8.9$ $[+4.0, +13.9]$ | $+4.6$ $[-0.1, +9.3]$ | .51 |
| MME Gate-0 | 196 | $+8.2$ $[+3.3, +13.2]$ | $+3.0$ $[-1.1, +7.2]$ | .37 |
| TextVQA held-out | 250 | $+7.9$ $[+4.2, +11.6]$ | $+4.4$ $[+1.1, +7.6]$ | .56 |
| TextVQA prospective | 600 | $+9.8$ $[+7.1, +12.6]$ | $+5.6$ $[+3.1, +8.2]$ | .57 |
| TextVQA v3 | 300 | $+11.4$ $[+8.0, +15.1]$ | $+6.7$ $[+4.0, +9.4]$ | .59 |
| TextVQA strong | 400 | $+9.4$ $[+6.3, +12.5]$ | $+6.3$ $[+3.6, +9.0]$ | .67 |
| TextVQA hidden | 400 | $+7.8$ $[+4.7, +10.9]$ | $+4.1$ $[+1.5, +6.8]$ | .53 |
| TextVQA Gate-0, Phi-4 | 200 | $+9.9$ $[+5.7, +13.9]$ | $+7.5$ $[+4.1, +10.8]$ | .76 |
| MME Gate-0, Phi-4 | 196 | $+2.3$ $[-2.9, +7.6]$ | $+1.5$ $[-3.0, +6.1]$ | .66 |

Restricting the oracle to the deployed family-level action space removes roughly 40–50% of the headroom on TextVQA (35–65% on MME); a family-level headroom of +4 to +7pp remains, with CIs excluding 0 on every fresh TextVQA set ($n \geq 250$). Budget-constrained (Lagrangian) values at 1/2× are .777–.781 (action-matched) versus .809–.819 (unrestricted) on the fresh sets. At the original 600-state 1/2× operating point, the unrestricted oracle reaches .818, the action-matched oracle .777, and the frozen selector .720: 4.1 points of the unrestricted 9.8-point oracle–selector gap are attributable to finer crop granularity, while 5.7 points remain between the action-matched counterfactual oracle and the realized selector. The opportunity therefore has two layers: roughly half is crop granularity (which box), which the selectivity audit shows is not question-predictable; the other half is family choice, which is inside the selectors' action space and remains largely unrealized (best deployable cell .749 vs. action-matched oracle .777 at 1/2× on the representation–head set). File: `deployable_oracle`.

## 9.14 Blinded human adjudication of scorer disagreement

**Sampling.** The adjudication universe consists of the 1,584 unique (state, normalized answer) pairs produced by the 13,200 generations of the 600-state prospective TextVQA evaluation. We sample 200 pairs for which the benchmark (strict) score is zero but the format-tolerant score is positive (245 such pairs exist), with allocation proportional to the dominant action family among the rollouts producing each pair (Crop 88, HighRes 49, OCR 42, Stop 21). We additionally sample 100 agreement controls: 50 pairs scored correct by both automatic scorers and 50 scored incorrect by both. Sampling is fixed by preregistration (seed 20260921) before any human label exists; the 300 pairs cover 1,527 rollouts, the 200 disagreement pairs 675.

**Blinding and adjudication.** Three independent annotators judge semantic correctness (correct / incorrect / ambiguous) from the image, the question, the reference answers with vote counts, and the model prediction only, without access to automatic scorer outputs, action arm, replay identity, or policy identity, under the instruction that containing the reference token is not sufficient for correctness. Majority vote defines the adjudicated label; a majority of "ambiguous" and pairs without a majority are retained separately.

**Agreement and semantic error.** Across the 300 sampled pairs, raw pairwise agreement is .906 (mean; Cohen's $\kappa$ .70 / .88 / .72), Fleiss' $\kappa = .766$, and Krippendorff's $\alpha = .767$; 292 pairs receive a majority label (258 unanimous), 7 are ambiguous by majority, 1 has no majority. Among the 197 resolved benchmark-negative, format-tolerant-positive pairs, 80.7% (Wilson 95% CI $[74.6, 85.6]$) are human-correct and 19.3% ($[14.4, 25.4]$) are human-incorrect. The human-incorrect share concentrates in partial-credit matches: 12.3% of pairs with format-tolerant score 1 ($n$=154) versus 43–46% at scores 1/3 and 2/3 ($n$=43). All 50 agreement-control pairs scored correct by both automatic scorers are human-correct; among the 45 resolved pairs scored incorrect by both, 11 are judged human-correct (paraphrases neither scorer accepts). Weighting the three strata by their population sizes (245 / 580 / 670), agreement with the human label is .758 [.699, .812] for the benchmark scorer and .856 [.797, .909] for the format-tolerant rule.

**Policy-level sensitivity analysis.** For every rollout whose (state, answer) pair was adjudicated with a majority label, the human label replaces the automatic score without reweighting (659 rollouts); all other rollouts retain their automatic scores. For unresolved or unsampled scorer-disagreement pairs, the pessimistic assumption retains the benchmark score of zero, whereas the optimistic assumption treats only format-tolerant full-credit matches as correct. All partial-credit matches, including scores of 1/3 and 2/3, remain incorrect under both assumptions. These are sensitivity bounds over unresolved scorer-disagreement cases, not formal bounds on semantic accuracy, because unadjudicated agreement cases remain automatically scored.

At the primary 1/2× operating point, frozen selector minus frozen mixture changes from +0.9 points ($[-1.6, +3.4]$) under benchmark scoring to +4.1 ($[+1.8, +6.5]$) and +4.2 ($[+2.0, +6.5]$) points under the pessimistic and optimistic assumptions; the format-tolerant estimate is +4.6 ($[+2.4, +6.8]$). At 1/4× / 1× the pessimistic values are +2.5 $[+0.2, +4.9]$ / +3.0 $[+0.5, +5.5]$ and the optimistic values +2.7 $[+0.4, +5.0]$ / +2.9 $[+0.5, +5.4]$. Only policies frozen before the 600-state set was collected are evaluated here. File: `human_adjudication`.

## 9.15 Cross-dataset and cross-reader transfer details

**Cross-dataset (TextVQA selector → VQAv2, zero adaptation).** Strict: parity with the cross-fitted mixture (−0.9/−2.7/−1.9pp, CIs include 0); the selector loses to the in-sample mixture at 1/2×/1× (that bound is baseline-favoring by construction). Lenient: directionally positive at every budget (+2.8/+2.0/+1.5pp, CIs include 0). The selector's action mix shifts sensibly (OCR-heavy on TextVQA → Crop/HighRes-heavy on VQAv2). Note: the oracle column in this analysis is the family-level allocation ceiling (4 arms, crop = uniform expectation) while mixtures draw on per-slot arms.

**Cross-reader (Qwen selector → Phi-4 states).** MME: the transferred selector nearly always stops (realized cost ≤73 tokens) and loses to the target-reader cross-fitted mixture by −8.0 to −10.4pp (CIs exclude 0), leaving Phi-4's headroom unrealized. TextVQA: the transferred selector beats always-Stop by +7.9 to +10.4pp (all CIs exclude 0) and is directionally ahead of the cross-fitted mixture (+1.3 to +2.7pp, CIs marginal). A confidence-recalibrated variant barely differs: the issue is the signal, not its calibration. Combined with the instance-level agreement analysis (help-set overlap ≈ chance on TextVQA; best-family agreement among both-help states 73–77%), what transfers is a coarse task-level signal of which action class helps, not instance-level perception need.

**Agreement conditional on baseline capability.** Because the readers' Stop accuracies differ widely, the pooled help-set overlap partly reflects capability rather than perception need. Stratifying the Gate-0 TextVQA states by whether each reader's Stop is correct: among both-Stop-wrong states ($n$=70 strict / 51 lenient) the Qwen–Phi-4 help Jaccard rises from the pooled .19 to .35 $[.23, .47]$ (lenient .64 $[.50, .77]$) and the correlation of per-state best-arm utility to .31 $[.14, .47]$ (lenient .48 $[.25, .69]$), while chance-corrected $\kappa$ stays ≈ .1 because both readers have high help rates there; the mixed stratum is anti-correlated (−.29 $[-.40, -.17]$). MME both-wrong: Jaccard .40, $\kappa$ .36 $[.17, .54]$. The instance-level claim is therefore "moderate overlap, weak beyond-base-rate agreement", not independence. File: `cross_reader_conditional`.

## 9.16 Latency at the token-matched operating points

Already-logged latencies of the same frozen operating points (never recomputed, never summed with tokens): the multi-action policy's low-token operating points substitute OCR and carry its tool time — TextVQA 1/4×: 29.0 s/state vs. 0.15 s for the confidence binary at the same token cost; VQAv2 1/4×: 11.1 s vs. 0.14 s; always-HighRes 0.67 s. On the latency axis the $\lambda_{\text{lat}}$ sweep avoids OCR and the binaries reach always-HighRes cheaply; OCR-heavy points sit 1–2 orders of magnitude to the right. Any efficiency claim must name its axis.

# 10 Infrastructure and Reproducibility Details

**Models and serving.** Qwen2.5-VL-7B-Instruct and Phi-4-multimodal-instruct, served with vLLM 0.11.0 (torch 2.8.0, CUDA 12.8), greedy decoding, fixed seed, logprobs enabled for STOP rollouts. Prefix caching is disabled — mandatory, because the protocol issues many near-identical requests per image and cached prefixes would couple arms that must be independent. Model weights are frozen throughout; no fine-tuning anywhere.

**Determinism and caching.** All generations are cached in SQLite keyed by model, prompt version, decoding parameters, and view bytes; changing any of these invalidates the cache rather than silently mixing configurations. STOP re-generation byte-match checks are asserted at collection time (0 mismatches on all reported collections; one disclosed exception: the sequential-v2 development confidence regeneration at 379/400).

**Scale.** In addition to the structural audits (7,000 rollouts per reader and dataset) and the held-out, lockbox, and answer-format collections, the prospective program includes the 600-state one-step selector evaluation (13,200 rollouts), the 300-state sequential evaluation (5,400), the 300-state hidden-state evaluation (6,600), the 400-state nonlinear-head evaluation (13,200, including a third replay), and the 400-state representation–head evaluation (8,800), plus 1,650 hidden-state extractions. In total the released logs hold about 117,000 model rollouts (excluding offline rescoring files and smoke runs). The human adjudication covers 300 unique state–answer pairs sampled from the 600-state prospective TextVQA collection. Every model-based evaluation collection passed the zero-transport-error integrity gate.

**Sampling and leakage control.** Every evaluation set is drawn with a fixed seed from a pool that excludes all previously used ids (a cumulative "burn list"); manifest SHA-256 prefixes are recorded in the corresponding preregistration document before collection and verified in the released manifests (the 600-state prospective manifest `a535d8edce3aa822`; the sequential 300-state manifest `4196e1abbead1506`; the v3 manifest `7bc5ec8a8eb9a61f`; the nonlinear-head 400-state manifest `7312457434be55f3`; the representation–head 400-state manifest `af41a4ac740a9ecb`; the adjudication sample mapping `1384f99672bc8d65`).

**Prospective-set separation and sequential development.** The one-step selector (600), sequential (300), hidden-state (300), nonlinear-head (400), and representation–head (400) prospective evaluation sets are mutually disjoint, disjoint from the Gate-0 development set (200, val split) and from the train-split development, lockbox, and held-out sets (250 each); pairwise overlap is zero for all nine sets. Each controller is evaluated only on a set that was never used for its own training or model selection. The research program is sequential rather than a collection of statistically independent replications: later controllers use previously analyzed sets in their training pools (1,350 and 1,750 states), but are evaluated on newly drawn untouched sets. Accordingly, the paper treats the preregistered primary endpoints as distinct prospective follow-ups and makes no familywise-confirmatory omnibus claim across them.

**Statistics.** State-level bootstrap (10,000 draws) over independent perceptual states; paired resampling for policy differences; no rollout-level resampling anywhere. Preregistered primary endpoints are single comparisons; all secondary comparisons are reported unadjusted and labeled as such.

# 11 Contents of the Supplementary ZIP

`supplementary.pdf`
: This document.

`code/`
: The complete audit framework and experiment scripts: `ppu/` (the library: budget accounting, actions, regions, OCR, prompts, replays, evaluator, cache, runner, audit), `scripts/` (every collection, analysis, and figure script named in the result files), `tests/` (offline tests exercising the full pipeline with stub

readers — no GPU or data required), and a README with environment and reproduction instructions. Absolute paths have been replaced by placeholders for anonymity.

`preregistrations/`
: All preregistration documents used in the study, verbatim, including the withdrawn original human-audit preregistration and the later preregistered prospective follow-ups (nonlinear-head, representation–head) and human-adjudication protocol.

`results/`
: Machine-readable result files (JSON) and their human-readable summaries (Markdown) for every analysis cited in the paper or in this document — including the action-matched oracle analyses, the human-adjudication summaries (blinded sample mapping, the three verdict files, and the analysis output), the nonlinear-head follow-up, the representation–head grid, the replay-pairwise and conditional cross-reader analyses — plus per-state CSV rows for the sequential prospective oracle and history analyses, and the frozen results digest.

`models/`
: Freeze artifacts for every learned component (pickle + JSON manifest with weights, scaler, confidence ECDF, feature list, and seed), matching the SHA-256 prefixes cited above.

`logs/`
: Complete raw rollout logs (gzipped JSONL) for every core experiment: the Gate-0 audits (both readers), the 600-state one-step prospective evaluation, the sequential development and 300-state prospective collections, the v3 prospective collection, the 400-state nonlinear-head and representation–head prospective collections, the held-out multi-action arms, the multi-replay robustness collections, the answer-format intervention arms, and the sealed binary-utility sets. Each row stores the full prompt metadata, served question, view labels and realized token costs, the *raw generation*, extracted answer, per-metric score, and (for STOP) token log-probabilities — every number in the paper is recomputable from these files without GPU inference, including under alternative scoring rules.

`manifests/`
: Evaluation-set manifests (sample ids, questions, references, native sizes, relative image paths) for all prospective, held-out, lockbox, and fresh sets (including the nonlinear-head and representation–head 400-state sets and the frozen 300-pair adjudication sample), verifying the preregistered SHA-256 prefixes. Images themselves are from the public TextVQA, VQAv2, and MME-RealWorld-Lite releases and are not redistributed.

**Reproduction.** Scoring, audit, and policy analyses are reproducible offline from `results/` and `manifests/` without GPU inference (raw generations are embedded in the result JSONs where needed; the evaluator is deterministic). Full re-collection requires serving the two readers with vLLM (scripts provided; prefix caching must remain disabled) and the public benchmark images. The offline test suite (`python -m pytest tests -q`) exercises every code path with stub readers and stub OCR.